

Carbon Footprint on Denim Manufacturing

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Abstract

Carbon footprint, also called carbon profile, defines the overall amount of carbon dioxide and other greenhouse gas (GHG) emissions associated with a product throughout the entire supply chain, from raw materials to end-of-life recovery and

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disposal. Electricity production in power plants, heating with fossil fuels, transport operations, other industrial and agricultural processes, among others, cause these emissions. Indicators such as the global warming potential (GWP) are used to quantify the carbon footprint. As defined by the Intergovernmental Panel on Climate Change, GWP is an indicator that reflects the relative effect of a GHG in terms of climate change considering a fixed time period, such as 100 years (GWP100). The GWP for different emissions can then be added together to give a single indicator that expresses the overall contribution of these emissions to climate change. Denim garments are the most popular clothing worldwide; many countries are taking measures to reduce carbon emissions through the use of organic cotton, environmentally friendly methods of manufacturing, and optimized denim manufacturing. Carbon credits have been adopted by denim manufacturers; this helps them to achieve overall goals of carbon emissions. This chapter explains the concept of carbon footprint in denim manufacturing and describes case studies, difficulties facing the industry, and optimized denim manufacturing techniques that reduce the carbon footprint.

Keywords

Carbon footprint · Carbon credit · Denim manufacturing · Environmental impacts · Global warming potential · Sustainability

Introduction

Carbon footprint is the total greenhouse gas (GHG) emissions caused directly or indirectly by an individual or organization during production of products [1]. It is mostly expressed as a carbon dioxide (CO₂) equivalent or tons of CO₂ [2]. When driving a car, the engine burns fuel, which creates a certain amount of CO₂; that amount depends on fuel consumption and distance travelled. Using heat to boil oil or using electricity, gas, or coal generates CO₂. The production of foods and goods also emits some quantities of CO₂. The carbon footprint is the sum of all CO₂ emissions induced by the activities in each time frame. CO₂ is calculated based on fuel consumption. The next is to add the CO₂ emission to the carbon footprint. Other greenhouse gases (GHGs) such as methane might be emitted and ozone can be depleted because of human activities. Other GHGs are also taken into account in the carbon footprint. They are converted to an amount of CO₂ and referred to as equivalent CO₂—an amount that would cause the same effects on global warming.

During the Industrial Revolution, textile manufacturers were neither conscious of nor bothered by the effects of textile production on the environment, and so they disposed of chemicals into nearby rivers and water bodies. As a result, environmentalism and social consciousness emerged. As a starting point, consumers started to think of new ways to dispose of, recycle, and reuse clothes. Consumers became aware of how clothing is produced and in turn of the impact of the process on the environment, which paved the way to a meticulous lifestyle. Companies reacted to this lifestyle by providing eco-labels to inform consumers that they care and are

concerned with sustainability. Over the past decade, environmental issues (e.g., global warming) have been spotlighted by global society, which has an impact on other aspects of manufacturing, too. Henceforth, manufacturers enlarged production of sustainable products rather than disposable ones. This chapter assesses the life cycle of denim jeans, as we know that denim is a noteworthy part of the fashion industry because of its popularity. According to a report from Cotton Inc., U.S. consumers generally own seven pairs of denim at one time [3]. Life cycle assessment is a detailed analysis of a product that elaborates the entire lifespan of a product, from fibers to retail. Consequently, this encompasses the production of raw materials and, in the case of denim, the development process for denim garments, logistics, their use; it ends with denim's reuse, recycling, and disposal. Depending on consumer behavior, a product's use phase and end-of-life disposal phase—critical stages in a product's life cycle—are modified. The environment affects each stage of the manufacturing phase of denim, such as energy and water usage, CO₂ emissions, and waste [4, 5].

Concept of the Carbon Footprint

The concept of the carbon footprint revolves around transportation and heating issues. The fabric gives off a large amount of CO₂ and other “footprint content.” Regarding U.S. Energy Information Administration, the textile industry is the fifth largest contributor to the carbon footprint content within many countries [6]. The textile industry is large and it produces many GHGs. Textile production is estimated to produce around 60 billion kg of fabric. The amount of energy and water needed to produce a boggle of fabric includes 1074 billion kWh of electricity or 132 million metric tons of coal and 6–9 trillion L of water. Textiles, including clothing, accounted for nearly 19.8 tons of total CO₂ emissions. Therefore it is necessary to study how to estimate the carbon footprint of a fabric—in this case, to consider the “embodied energy” of the fabric—because that gives an idea about the energy consumption during each process. To estimate the embodied energy, it is important to know the type of fibers used to make the fabric, because that gives an idea about the energy required to manufacture the fibers and yarn. In general, the carbon footprint varies a lot for different types of fibers, so the energy required to produce them also varies. Next, we have to determine the energy required to spin the yarn into a fabric, because once a fiber has been converted to filament, the amount of energy needed to convert the filament into a fabric is consistent. Many factors must be considered for natural fibers: preparation of the field in which the plant was grown, weed control, pesticide use, and harvesting method; these are major factors in determining cost. Just 1 ton of nitrogen fertilizer emits about 7 tons of CO₂. Synthetic fibers are mainly made from fossil fuels, and a very large amount of an energy is needed to take oil from the ground; the energy needed to create synthetic fiber is much higher than that needed to create natural fiber [6–8]. However, denim is made from many textile fibers—namely, cotton, polyester, lyocell, and polyurethane. Choosing natural fibers has two major benefits:

- They can be easily degraded by microorganism and by other attacks, so the CO₂ content will be released, whereas man-made fibers do not easily release content; instead, they release some heavy metals, recycling them requires a high cost, and an estimated 3 tons of CO₂ emissions produce 1 ton of material to be burned.
- Sequestering carbons is the process by which CO₂ in the upper atmosphere is absorbed by plants on the Earth and is stored as carbon in biomass. For example, jute absorbs 2.8 tons of carbon/year.

Substituting organic fibers for conventionally grown fibers is little better because it only uses less energy for production, has fewer costs, and emits a slightly smaller amount of CO₂ gas. A study found that natural fibers emit 43% less GHG than the conventionally grown fibers, and organic fibers use 63% of energy used for farming, which is much less than that used for farming conventional fibers. Another study also stated that in long-term controlled trials, organic farming gave about 100–400 tons of carbon/year; this helps to reduce GHGs present in the environment [13]. The main thing lies in handling the organic matter; organic matter in soil is based on carbon, and increasing the amount of organic matter directly relates to carbon sequestration. Organic farming also helps to reduce environmental impacts and enhance social responsibility. Organic farming can eliminate the use of synthetic fertilizers and pesticides, which cause series problems, and can save larges amounts of water, which can be used instead for irrigation. It also ensures a safe mode of living. Organic agriculture is being closely examined; studies found that it is a climate change tool that can be used to fight against global warming. The energy required to weave different types of fiber into fabric is same, since the processing method is same for nylon, wool, cotton, and polyester [6, 7].

Evaluation of the Carbon Footprint

A significant phase in determining the denim life cycle (see Fig. 1) is GHG emission; laundering and drying are the main determinants the life cycle of a garment. Maximal energy is required to wash, dry, and iron denim garments. Throughout the total life cycle, laundry accounts for 40–50% of GHG emissions due to energy inputs. Machine drying is among the highest energy users producing GHG emissions. The use of renewable or nuclear power for laundering reduces GHG emissions in a garment's life cycle. Hand-washing and dry cleaning require very little energy and therefore result in few GHG emissions. GHG emissions can change according to the region and the customer's behavior. Studies of denim garments reveal that laundry is done more frequently than is necessary, which increases total GHG emissions [6, 9, 10].

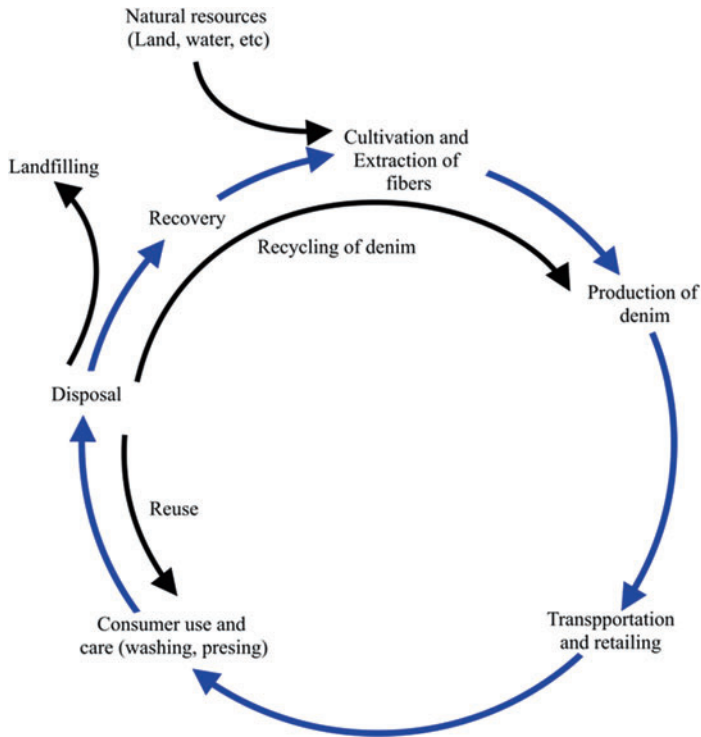


Fig. 1 Phases of a denim life cycle (reprinted from Periyasamy et al. [6] with the kind permission of Elsevier Publication)

Product Life Cycle

Raw Material Procurement

Cotton is the main raw material for denim [11]. At times, synthetic materials are used to regulate additional properties such as comfortable fit and durability. One of the most accepted natural fibers used worldwide is cotton. A total of 3% of cultivated land is used to produce cotton. According to a previous study, natural fibers are environmental friendly, unlike synthetic fibers [12]. Each fiber is different, with its own sustainability challenges in the production process that cannot be disregarded. In general, cotton is a natural fiber that is grown, not manufactured (see Fig. 2). Pesticides and fertilizers are required to achieve the expected quality of cotton, therefore, even though it is environmentally friendly, it also consumes a large quantity of water [13].

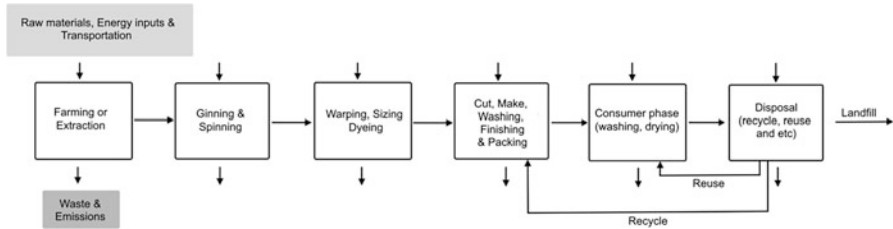


Fig. 2 Life cycle analysis studies of denim made with natural fibers.

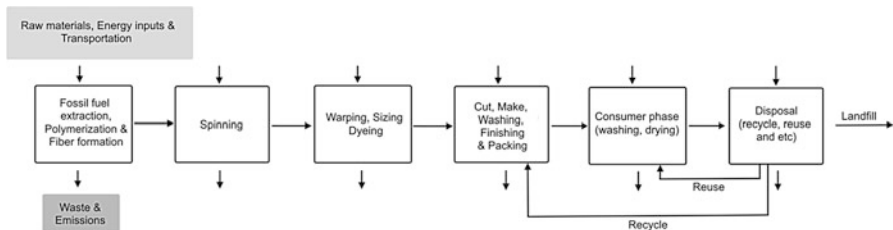


Fig. 3 Life cycle analysis studies of denim made with synthetic fibers

Based on a data sheet published by the U.S. Department of Agriculture and the Organic Consumers Association; cotton is the most toxic crop in the world. Cotton uses more than 25% of all the insecticides and 12% of all the pesticides applied worldwide [14, 15]. The majority of these insecticides are considered to be the most toxic chemicals in the world. A total of 7–29 tons of water are required to produce 1 kg of raw cotton [16–18]. It also contributes to global warming potential (GWP), not only with CO_2 but also other gases such as N_2O and CH_4 . GWP for cotton cultivation is 53% of CO_2 , 45% of N_2O , and 2% of CH_4 [19]. Because of the aforementioned environmental and health issues, most denim brands are committing to using organic cotton materials [20, 21]. As for synthetic materials, polyester and polyurethane are the most used in denim manufacturing. Although the manufacturing of these fibers requires less water and land, it has other effects on the environment. In general, synthetic polymers are obtained from nonrenewable resources (e. g., fossil fuels), and the production of these fibers requires an enormous amount of energy, causing higher GHG emissions (see Fig. 3). Environmental problems in waste management, such as nonbiodegradability, pose various health and toxicological threats that are also negative impacts of the production of synthetic fibers [22].

Manufacturing

Manufacturing is the process whereby textile fibers are converted into yarn, then into the fabric, and finally into garments (three-dimensional structures). Figure 4 shows a

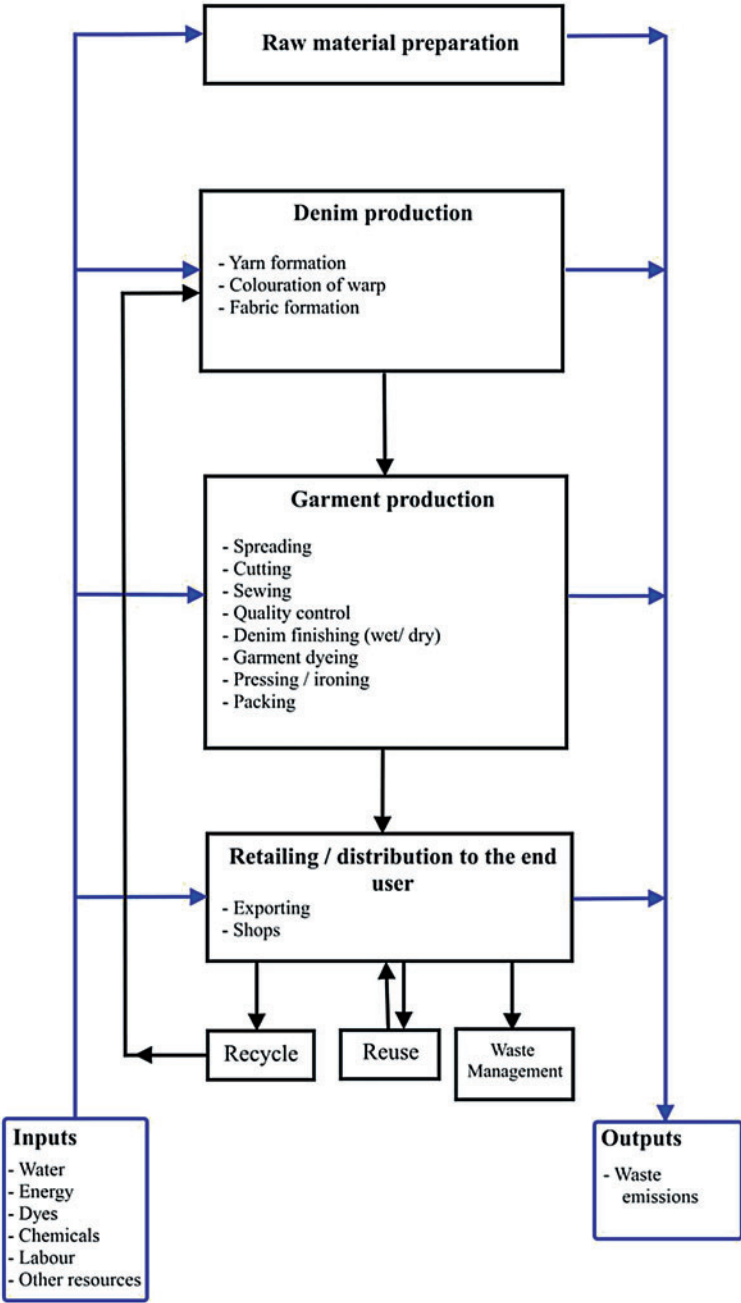


Fig. 4 Production life cycle for denim (Reprinted from Periyasamy et al. [6] with the kind permission of Elsevier Publication)

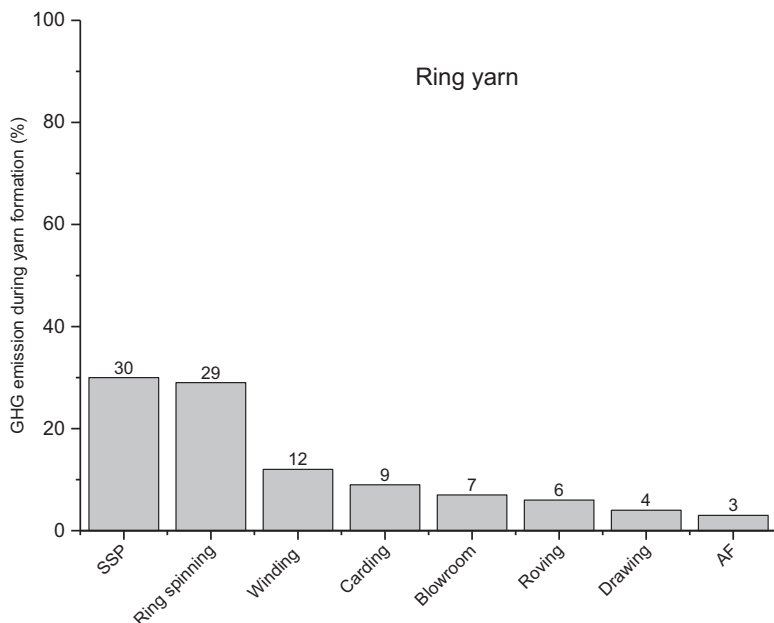


Fig. 5 Greenhouse gas emissions during ring yarn manufacturing. *SSP* supplementary spinning processes, *AF* administrative facilities

map of each process, which is explained in relation to denim manufacturing with respect to the inputs and outputs (emission of waste) of the production chain for a product's life cycle.

Spinning

In general, different process (e.g., opening, carding, combing, drawing, roving, spinning, and winding) are carried out using cotton and other fibers to form a yarn, requiring a large amount of energy [22, 23]. The results of the study conducted by us of GHG emissions with respect to yarn manufacturing from blowroom to yarn winding from the spinning industry in Coimbatore, India, are shown in Figs. 5 and 6. Among the process, ring spinning machine, rotor spinning machine and winding machine consume the largest energy, which is nothing but largest GHG emission process/machines.

Coloration of Denim and Fabric Formation

Denim fabrics have twill structures obtained through a weaving process (see Fig. 7). Whenever the process of warping is completed, the warp sheets must undergo a

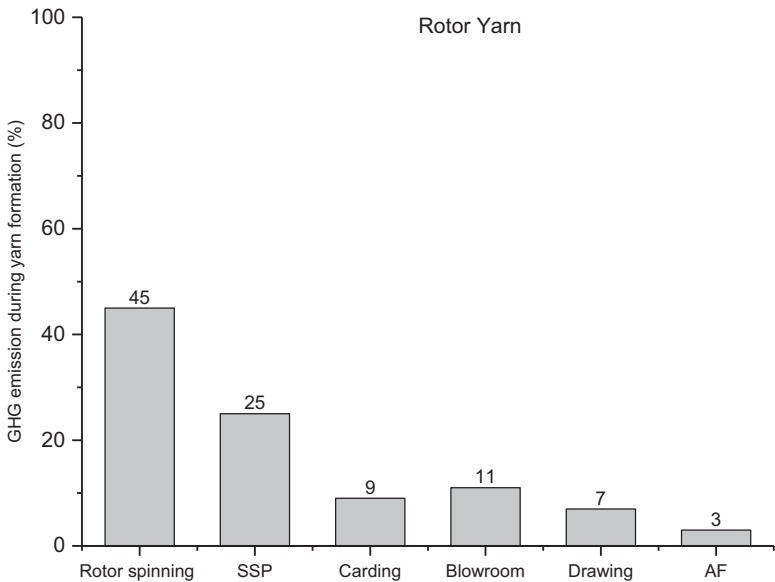


Fig. 6 Greenhouse gas emissions during rotor formation. *SSP* supplementary spinning processes, *AF* administrative facilities

Fig. 7 Typical denim garments



sizing process; the sizing box contains natural or synthetic sizing agents such as polyvinyl alcohol or carboxymethyl cellulose [24, 25]. The temperature is high during the process of coloration. Hence, carbon is engendered during this process through the generation of steam. A GHG emission study was conducted in the India-based denim industry, and the results are plotted in Fig. 8, which shows the GHG emissions during the weaving preparatory process. Among the various process in

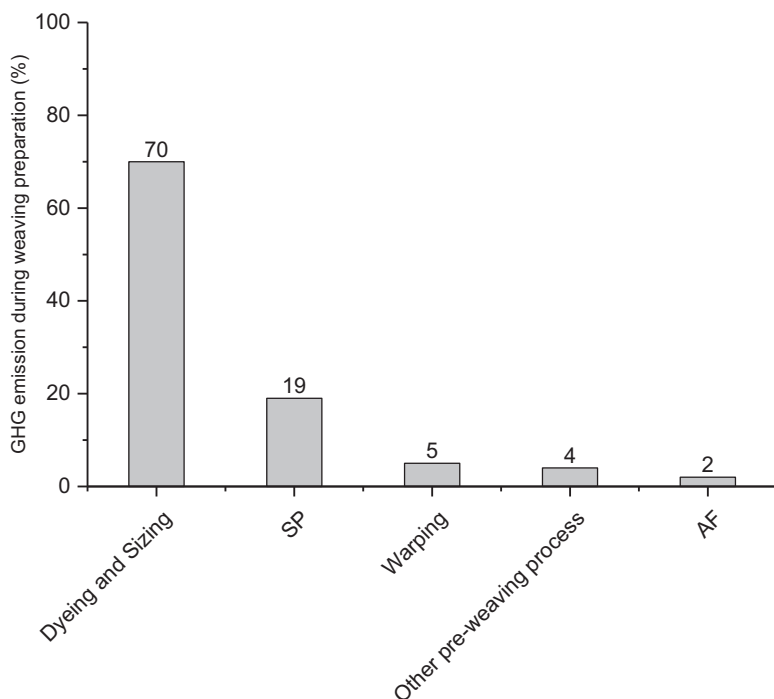


Fig. 8 Greenhouse gas emissions from the weaving preparation department with respect to different operations

this phase, dyeing and sizing supply 70%; supplementary process (air compression, lighting) cause 19%; warping, 5%; and other preweaving processes such as drawing in and denting, 4%. As far as denim fabrics are concerned, only the warp yarns are dyed. The warp sheets of denim are generally colored with indigo and sulfur dyes. The excess dye is removed only after washing it in water after dyeing, which causes water pollution and wastewater. During indigo dyeing, sodium hydrosulfite is added to reduce the indigo dyes is nothing but, this chemical reaction makes indigo to the soluble form, therefore, it is possible to carry out the dyeing process [25–28]. The reduced indigo dyes are normally yellow; it turns to blue when exposed to atmospheric air as a result of oxidation [29]. Therefore the dyeing and sizing processes require steam energy, which contributes 96% of GHG emissions of the total emissions in this phase.

GHG emissions during the weaving process (for denim fabric) were studied in the denim weaving industry located near Coimbatore, India. The results are plotted in Fig. 9, showing that the weaving machines, air compressor, and air conditioners contribute 85% of GHG emissions. However, the air compressor alone contributes 29% of GHG emissions; in general, compressed air is the heart of air-jet looms. However, these emissions can be reduced to some extent by properly maintaining the machines.

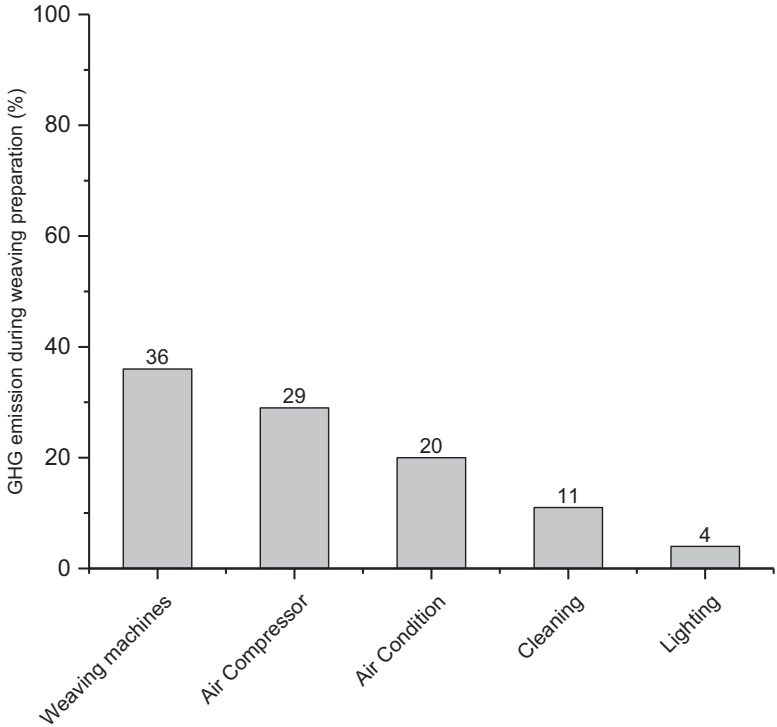


Fig. 9 Greenhouse gas emissions during the weaving process

Garmenting and Washing

“Garment manufacturing,” also known as the cut-make-trim stage, is the final stage in the denim supply chain. Various operations such as spreading, cutting, sewing, heat pressing, ironing, and packaging are involved in this process. An important aspect in the denim supply chain is the logistical activities during the production stages, from raw materials to finished denim, which are usually situated in diverse places and countries. Transportation of goods is not only involved in manufacturing; when a product is completed it has to be sent to wholesalers, retailers, distribution centers, and shops using different kinds of transportation (e.g., by road, ship, or air) that all consume various amounts of energy and emit various levels of CO₂. Freight transport is estimated to produce 8% of CO₂ emissions. In reference to LCA, transportation is a massive factor in products reaching consumers, and it consumes a huge amount of oil and gas, the prices of which increase progressively, thereby increasing transport costs. Diminishing the amount and distance of transport of goods has a significant effect; this is popularly known as “green logistics,” which ensures the management of a green supply chain [30–33].

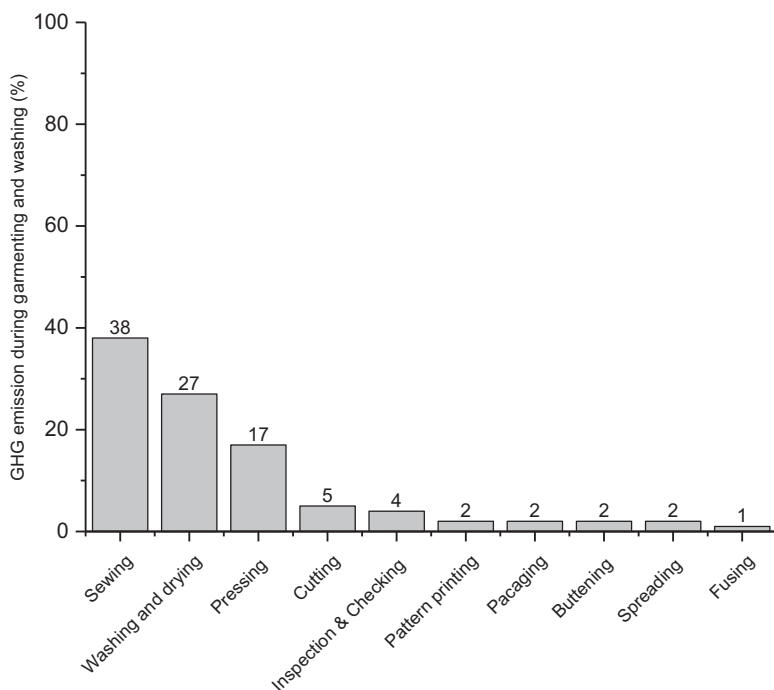


Fig. 10 Greenhouse gas emissions of the garmenting and further washing process

Based on the results from the garmenting process presented in Fig. 10, sewing, washing, and pressing consume the largest amounts of energy, together contributing about 82% of total GHG emissions. However, the washing/drying process includes the compressor inputs, belly washing machine, steam generation, and administrative facilities, whereas the sewing process includes illumination, compressors, and administrative facilities.

Consumer Use and Disposal

Today, well-meaning people who truly want to protect the environment can be passionate about what should be done with waste, which they may consider to be a generic material. Use of the LCA process is geared toward the consumer and is no longer within the manufacturer's control (e.g., passing valued garments on to family, friends, or charity; re-selling or recycling) and varies among and within regions and cultures. This phase encourages consumers to buy only when needed and to take care of clothes to prolong their lifespan. However, most denim clothing is washed in water every 2 weeks and then disposed of in a landfill after 1 year. Most companies put much emphasis on the care of denim. The main motto at the disposal stage—ensuring recycling and donating in order to reduce the amount of waste going to

landfills—helps to positively affect the environment. The most commonly known waste management strategies are the 3 Rs: reduce, reuse, and recycle; these generally save energy and money, reduce GHG emissions and global climate change, and, most important, helps to sustain the environment for future generations. The meaning of *recycle* is to collect and process materials that are considered trash and turn them into new products. Recycling saves resources and uses less energy than does the production of new items. A few companies have recently extracted fiber from old denim (considered trash) for technical applications such as insulation for homes and civic buildings [6].

Calculating the Carbon Footprint

To calculate the carbon footprint, several data have been considered, for example, the type and origin of the fiber; fiber loss during ginning, spinning, and other process; type of weave structure; weaving machines; sizing (add on a percentage); type of dyes; depth of shade; desizing (percentage loss), marker efficiency during the cutting process; sundry efficiency; chemicals used in and type of washing; weight of the packaging material; and mode of transport with respect to the entire supply chain.

Functional Unit

The functional units are the items to be analyzed in a study. Therefore, for this study the functional unit is the production of one pair of jeans with a medium stonewash, weighing 570 g. The details of the functional units are listed in Table 1.

System Boundaries for the Carbon Footprint

The life cycle of denim jeans from the beginning to the end includes raw materials extraction, fabric formation, coloration, garment manufacturing, transportation, retailing, and distribution.

Production Phase

In the production process, the raw material (fiber) undergoes many processes during the conversion to finished product. The number of processes during the production phase affects the impact on the environment; more processes leads to a larger effect. The different processes involved in making the final products are, for example, yarn formation, fabric formation, coloration, garment manufacturing, and the finishing process.

Table 1 Functional units for carbon footprint analysis

Functional unit	Men's bottom (36-inch waist)
Material	98% cotton, 2% Lycra
Raw material	Cotton (from India)
Dye	Indigo blue
Finishing	Medium stone wash with 3-D finish
Weight of jeans	570 g (~ 20 oz)
Year of production	2015
LCA conducted	2016
Consumer phase	India
ISO	ISO 14040

ISO International Organization for Standardization, *LCA* life cycle analysis

Climate Change

One of the major concerns of twenty-first century is the change in climatic conditions. The textile industry is a major polluting industry that has drawn attention to changes in climatic conditions, which have resulted in global warming, increased sea levels, and drastic weather conditions. Increases in the consumption of clothing (i.e., textiles) have greatly affected climate changes. Throughout the production of denim, from the beginning to the finished product, a large amount of carbon is emitted, which initiates climate change. These changes will affect human health and the ecosystem, and they increase year by year. This study focuses on the GWP and measures the emission of CO₂ and other GHGs into the environment. GHG emission is inevitable in textile production; as a result it increases the system temperature (i.e., ecosystem), causes health hazards to humans and animals, and damages the entire ecosystem. Changes in climatic conditions with respect to the different phase of denim production (expressed as kilograms and percentages) are shown in Fig. 11. Consumers have the greatest impact on climate change; GHG emissions may increase when consumers use warm water to wash clothes and use a dryer to dry them; this affects climatic conditions. However, these can be reduced when clothes are washed with cold water and the dryer is used less often. Therefore, increased CO₂ emissions are due to factors such as washing with warm water, using a dryer, washing weekly, and using a conventional washing machine.

In general, washing only after wearing 10 times is a suggestion to strongly reduce carbon emissions. However, in order to heat the drum to dry jeans, the dryer must use steam or gas, and a huge amount of energy is required for this purpose. This is commonly known as the latent heat of vaporization. For example, wet jeans weighing 2 kg after washing require 1.4 kWh of energy to dry [34–36].

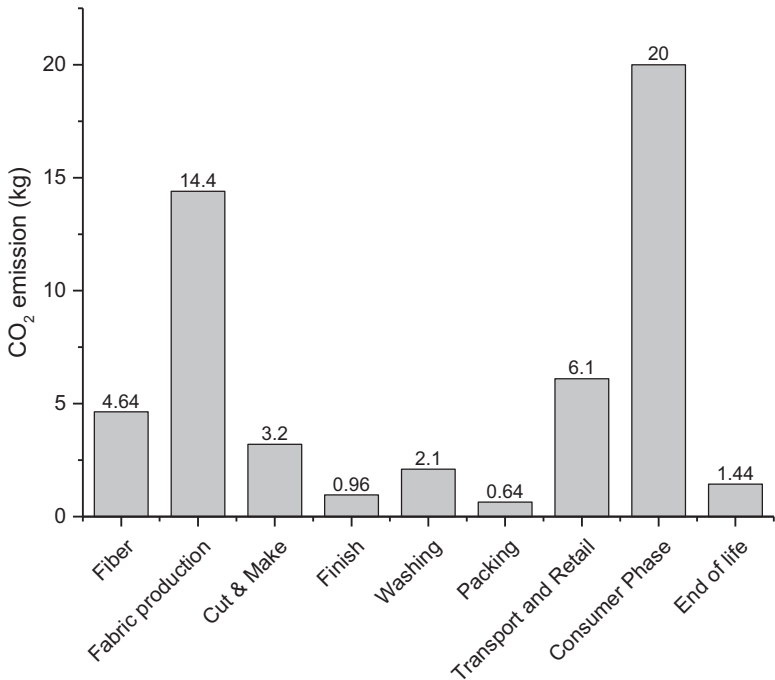


Fig. 11 Climate changes with respect to the different phases of denim production. *Fiber* cultivation to harvesting for cotton and chips for fiber for Lycra; *fabric production* spinning, sizing, dyeing, and weaving

Interpretation and Conclusion of Results

The main focus of an environmental analysis is the LCA. As restrictive eco-standards are introduced more often, the carbon footprint could be one of the criteria used to evaluate the environmental changes and caused by the global warming and cooling potentials. The main aim of this chapter was to assess the carbon footprint of any product, calculating the energy used to process them from yarn to garment. In producing a pair of jeans, the fabric production and consumer care stages have the largest impact on climatic conditions, because of the number of processes that use resources. Water bodies create nutrient-rich runoff water, which is damaged by the use of fertilizers and pesticides, which generally affects the oxygen content in the water, which thereby affects aquatic systems.

Challenges

This study identifies hidden facts about the production of denim as they relate to the carbon footprint and manufacturing. It drives choices to the consumer, who must choose products based on their environmental impacts. However, this study can also help to improve consumer awareness. The promotion of sustainable denim is a key position of retailers—hence the increase in eco-design depends on consumer awareness and preferences. When compared with rural people, urban people are more aware of the environmental impact of their clothing [6, 7]. When we discuss environmental impacts with denim manufacturers, they suggest that the consumer phase of the denim life cycle has the biggest impact on the carbon footprint. Hence it is necessary to educate consumers regarding the selection of the clothes they wear every day; this will change the environmental impact of denim [7, 10, 23, 35].

Conclusion and Further Outlook

The carbon footprints associated with large universities have a major impact on the environment for our society. Globalization is inevitable and unavoidable in the present world economic situation. Many industries are affected positively or negatively by the globalization trend. Sustainability has become an essential attribute of today's textile and clothing industry, the process of transforming the textile industry into more sustainable one is very sensitive, needs a lot of knowledge, skills and commitment. Sustainable apparel products can be defined as a part of the growing design philosophy and trend of sustainability, the goal of which is to create a system which can be supported indefinitely in terms of environmentalism and economic responsibility. Conventional textile production is one of the most polluting industries on the planet. The World Bank estimates that the textile industry is responsible for as much as 20 percent of industrial pollution in our rivers and land. Environmental impacts occur throughout the production chain and most attention is paid to water consumption, chemical use, and waste and greenhouse gas emissions. The textile industry has taken several initiatives to reduce its environmental impact and improve economic conditions.

The textile and apparel industries are notorious for their excessive contribution of waste and pollution to our environment. The unsustainable practices trends of poorly designed, raw materials selection, processing, manufacturing, and disposing of apparel product is a major contributing factor to this problem. The responsibility for ensuring sustainable work environment is reflecting from all most sides i.e., producers, consumer, buyer, government, nongovernmental agencies. The ongoing global competition for cleaner and lower price, women work environment and wage have increased to the satisfactory level for some countries. Hence, growing awareness and leading them towards sustainable practices in the textile and clothing business for a greener world. Part of the problem is that in today's complex global supply chains, where clothing is produced worldwide, the consumer has become further and further removed from producers and the origin of products. Despite the

textile and fashion industry's challenges, fashion can become more sustainable and people involved in the apparel supply chain can benefit. But in order to create an economically sustainable fashion industry we need to rethink the way we design, produce, consume and dispose of fashion, we need to redress fashion, so to speak. The solution to the problem lies in understanding the real costs of producing fashion.

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